Name _____ Period _____ Date _____

6-3 Lesson Quiz: Logarithms

short form

Questions keep their Savvas Lesson Quiz numbers (1, 3, 5), the numbers your packet uses.

1. Which function is the inverse of the exponential function $y = 3^x$?

Ⓐ $y = x^3$ Ⓑ $y = 3^x$ Ⓒ $y = \log_3 x$ Ⓓ $y = \log_x 3$

Explain your choice. Use the words *base*, *exponent* and *inverse*.

3. Which is the logarithmic equation $\log_8 64 = a$, written in exponential form?

Ⓐ $64^a = 8$ Ⓑ $a^8 = 64$ Ⓒ $a^{64} = 8$ Ⓓ $8^a = 64$

5. What is the solution to the equation $\log(2x + 4) = 2$? Round to the nearest thousandth, if necessary.

Show your work.

$x =$ _____

Scoring: Question 1 and Question 5 are 10 points each. Question 3 is a bonus of 5 points. Total out of 20.

6-3 Lesson Quiz, short form: answer key

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1. C, $y = \log_3 x$**.** The logarithmic function $y = \log_b x$ is the inverse of the exponential function $y = b^x$; here the base is 3. ($\log_3 x$ is the exponent you raise the base 3 to in order to get x, so it undoes $y = 3^x$.) A is a power function (the variable is the base, not the exponent); B is the function itself; D swaps the base and the argument. Savvas quiz item 1, DOK 2, skill F69. Example 1 (continued), Try It 1 and the packet Practice. **10 points:** 7 for C; 3 for an explanation that names the base 3 and says the logarithm undoes (is the inverse of) raising 3 to an exponent.

3. D, $8^a = 64$**.** The base 8 raised to the exponent a gives the argument 64 (and $a = 2$). Distractor A swaps base and argument; B and C put a in the base. Savvas quiz item 3, DOK 2, skill F73. Taught by the form switch in Example 5 Part B and Try It 5a (and the packet's top box). **Bonus 5:** D.

5. $x = 48$**.** Exponential form: $10^2 = 2x + 4$, so $100 = 2x + 4$, $2x = 96$, $x = 48$. Check: $\log(2 \cdot 48 + 4) = \log 100 = 2$. Savvas quiz item 5, DOK 2, skill F69. Example 5 Part B and Try It 5a. **10 points:** 4 for the exponential form $10^2 = 2x + 4$ (or $100 = 2x + 4$); 6 for $x = 48$. A bare 48 with no exponential form earns 6 of 10.

Items left off this form (available for a retake or Form B): Savvas item 2, the best estimate of $\ln 16$ (answer B, 2.77; DOK 2, F73; Example 4, cut; $\ln 16 \approx 2.7726$ by calculator); Savvas item 4, select all the logarithms evaluated correctly (answer A and E: $\ln 1 = 0$, $\log_5 \frac{1}{125} = -3$; DOK 1, F69; Example 3, cut).

IXL replacement: SmartScore 70 or above on 8RK and BXU replaces this quiz with 20/20.